University of BRISTOL

EEMEM00006 Analogue Integrated Circuits

Coursework Brief 2026/2027 | TB1

1 Introduction

This document outlines the coursework for the EEMEM0006 Analogue Integrated Circuits unit. The assessment is designed to evaluate your ability to apply analogue integrated circuit design principles to realistic engineering challenges. You will be required to integrate theoretical knowledge with transistor-level circuit design, validate performance through simulation, and critically evaluate design trade-offs, thereby demonstrating achievement of the module learning outcomes.

This assignment comprises two design tasks, which must be completed in pairs. A single report covering both tasks and all supporting design files required by this brief must be **submitted via Blackboard by 1:00 pm on 1 December 2026.**

1.1 What should be submitted

Task	Items to Submit
Design Task 1: Power Management Unit (PMU) Design [Max. 60 Marks]	☐ Technical Report (Max. 08 pages) ☐ Cadence Virtuoso Design Files
Design Task 2: Surface EMG Recording Amplifier Design [Max. 40 Marks]	☐ Technical Report (Max. 08 pages)

Each design task section outlines the expectation for each student to take responsibility for a clearly defined subsystem or aspect of the design. Individual marks will be awarded based on demonstrated personal contribution, technical understanding, and critical reflection. **Credit will be awarded for sound engineering methodology, justified design decisions, and rigorous analysis even where some specifications are not fully achieved.** The report accounts for 100% of the unit assessment, is marked on the 20-point scale (See Appendix A), and final individual marks will be adjusted based on actual contribution levels. Formatted in 11-point Arial with single line spacing and 2 cm margins, the report will be 16 pages in total, and one member from the group should submit it.

Individual Contribution Statement: The report must clearly identify the work undertaken by each student. Contributions should be linked to specific design activities, calculations, simulations, design decisions, and verification tasks. Individual marks may be adjusted where there is evidence of significantly unequal contributions. Where contribution levels cannot be adequately determined from the report and project records, an individual viva may be conducted to assess technical understanding and contribution.

1.2 Use of AI in Assessment

In this assessment, students may make use of Category 2 (Minimal) generative AI tools, as defined in the University of Bristol's guidance on using AI in assessment [1]. Under this category, students are permitted to use AI only for limited, surface-level support such as spell-checking and grammar-checking, but not for rewriting, generating, or substantially editing content. This approach aligns with the University's default position on AI use and helps maintain academic integrity by ensuring that students' submitted work continues to demonstrate their own understanding and skills. Clear adherence to Category 2 ensures that learning outcomes are met while still allowing students to benefit from basic assistive technologies

[1] https://www.bristol.ac.uk/students/support/academic-advice/using-artificial-intelligence/

Note: *All students within a group are expected to contribute and engage with the project throughout the duration of the assessment to meet all learning outcomes required for accreditation. Students will report and be monitored regularly on individual contributions, which will be used to award individual marks if the contribution of team members is not even. Where there is insufficient evidence to confirm an individual's contribution and/or that they have attained all of the intended learning outcomes for the unit, an individual viva may be required to provide further evidence of contribution and attainment.*

1.3 Learning Outcome Mapping

Learning Outcome	**AHEP 4 Mapping (BEng/MEng)**
LO1: Physics of Integrated Devices: Describe and explain the physics of integrated devices in terms of the mechanisms involved on a semiconductor platform.	C1 / M1; C2 / M2
LO2: Analogue Circuit Design and Performance: Analyse how analogue circuit design decisions influence device- and system-level performance metrics.	C1 / M1; C2 / M2; C3 / M3
3. Circuit Topologies and Signal Processing: Design and apply appropriate integrated circuit topologies to achieve the required analogue signal processing functionality.	C3 / M3; C5 / M5
4. Teamwork and Reflection: Function effectively as an individual, and as a member or leader of a team. Individually evaluate the effectiveness of your own and of your team's performance	C16 / M16; C17 / M17

2 Design Task 1: Power Management Unit (PMU) Design

2.1 Background

BM&M, a leading microprocessor design company based in Bristol, is currently developing a new line of energy-efficient microprocessors for next-generation of devices. To meet their strict power budget and noise-immunity requirements, the hardware team requires a dedicated Power Management Unit (PMU) for the microprocessor unit. The PMU serves as the heart of the microprocessor's power distribution network, ensuring a stable, independent, and clean supply voltage despite variations in temperature, process corners, and dynamic load currents.

In this coursework, your team will step into the role of an Analog IC Design Engineer tasked with designing, simulating, and verifying this critical PMU sub-system from the transistor level up.

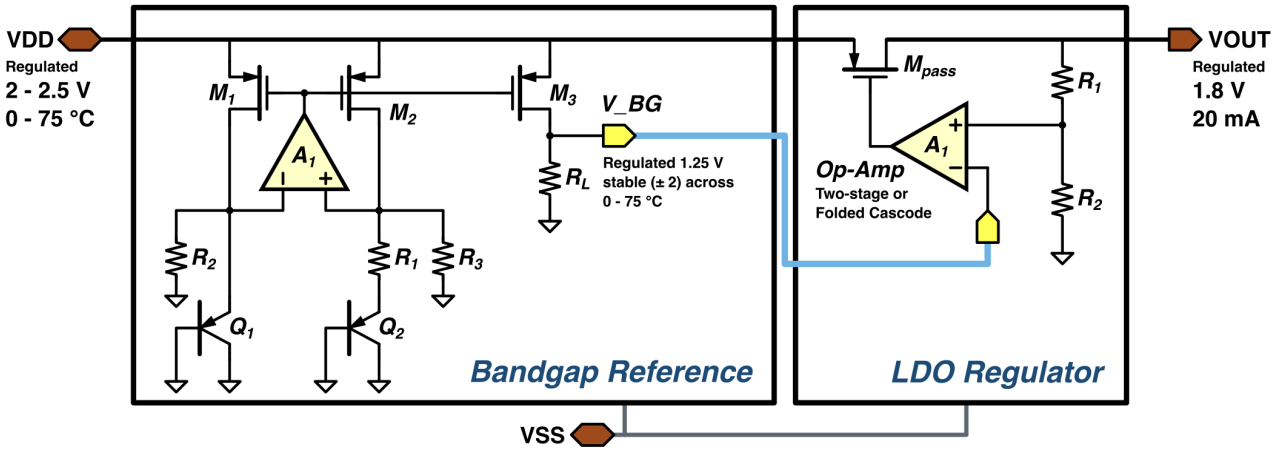


Figure 1: PMU Block Diagram

A typical PMU consists of two major blocks: a Bandgap Reference (BGR) followed by a Low-Dropout Regulator (LDO) as shown in Figure 1. Each of these units requires an operational amplifier for feedback control and error amplification.

Figure 2 shows a recommended op-amp architecture for use in the LDO and BGR designs. Students may either use this architecture or select an alternative op-amp topology, provided that they justify their choice and demonstrate that the design satisfies the required specifications (See Table 2).

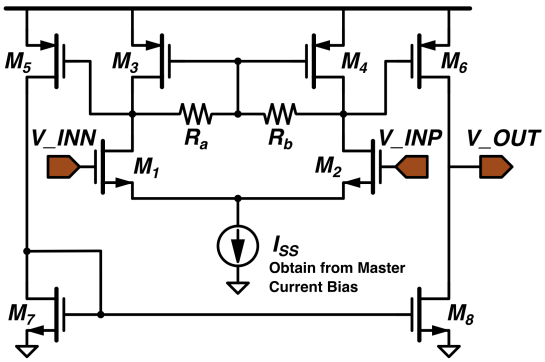


Figure 2: Two stage Operational Amplifier Circuit

The PMU should be implemented using GPDK180 at transistor level in Cadence Virtuoso and verified using simulation. All the design files should be submitted with the report for verification.

The PMU shall provide a 1.8 V regulated output at load currents up to 20 mA. The BGR shall provide a nominal 1.25 V reference, with a maximum-to-minimum output variation below 2 mV over 0–75 °C. The LDO loop phase margin shall be at least 60°. Table 1 defines the numerical requirements.

For each simulation, state the supply voltage, load current, output capacitance and Equivalent Series Resistance (ESR), reference loading, and process/temperature conditions. Provide reproducible DC, supply/load-sweep, startup, transient, loop-stability, Power Supply Rejection Ratio (PSSR) and noise simulations. Include amplifier, bias and startup-circuit currents in the corresponding block's quiescent-current budget, excluding the external load current; identify any shared bias current and count it once in the total PMU current.

Table 1: Target Specifications for the PMU

Parameter	**Target Specification**
LDO input voltage	2.0 - 2.5 V
LDO output voltage	1.8 V
LDO maximum output current	20 mA
LDO quiescent current	$< 100\,\mu$A
LDO dropout voltage	< 200 mV
LDO PSRR	≥ 40 dB at 100 kHz
LDO integrated output noise	$< 100\,\mu V_{RMS}$, integrated from 0.1 Hz to 100 Hz
BGR supply voltage	2.5 V maximum
BGR output voltage	1.25 V
BGR output voltage variation	Maximum-to-minimum V_{REF} variation < 2 mV over $0-75^{\circ}$C
BGR PSRR	≥ 40 dB at 100 kHz
BGR quiescent current	$< 100\,\mu$A

Table 2: Operational-Amplifier Design and Verification Requirements

Parameter	**Design and Verification Requirement**
Implementation	Use permitted GPDK180 devices. Justify the selected topology and explain any differences between the BGR and LDO amplifiers.
Supply and startup	Use the corresponding block's supply arrangement. Verify correct startup behaviour without relying on an already-established regulated output.
Input common-mode range	Demonstrate that the amplifier input common-mode range covers the operating voltages of the BGR or LDO under the assessed operating conditions.
Output swing and drive	Verify that the amplifier can drive the actual connected load, including the LDO pass-transistor gate, across the assessed operating range.
DC gain	Report the open-loop DC gain at the relevant operating point and relate the achieved gain to reference-voltage accuracy and output-voltage regulation.
Bandwidth and slew rate	Report the unity-gain bandwidth and slew rate under the specified loading conditions. Justify the achieved values using startup and transient-response results.
Stability	Verify stability in the actual feedback configurations. For the LDO, the phase margin shall be $\geq 60^{\circ}$. Report corresponding loop-stability results for the BGR.
Current consumption	Include the amplifier and all associated bias circuitry within the corresponding complete block's quiescent-current budget of $< 100\,\mu$A.
Noise and supply rejection	Assess and include amplifier contributions in the overall noise and PSRR verification of the corresponding block.
Verification evidence	State the supply voltage, input common-mode voltage, output operating point, load conditions, process corner, and temperature conditions. Submit reproducible simulation results.

2.2 Collaborative Work Strategy for The Group

To efficiently design the Power Management Unit (PMU) comprising the BGR, LDO, and Operational Amplifiers, partners can split the responsibilities as follows:

- **Joint Responsibilities:** Hand calculations for initial transistor dimensions (W/L), schematic capture,

cross-checking simulations, and final report editing; Design and verify the operational amplifiers using the selected topology or topologies; full system integration (connecting the BGR output V_{ref} into the LDO input) and compiles the final report.

- **Group Member A (BGR)**: Leads the design of the BGR block (utilizing op-amp A_1, bipolar transistors Q_1,Q_2, and startup logic shown in the schematic); Runs temperature coefficient (TC) and line regulation simulations for the BGR.
- **Group Member B (LDO)**: Focuses on designing and sizing the LDO power pass transistor (M_pass) and feedback resistor network (R_1,R_2); Performs AC stability (Phase Margin/GBW) and load transient response simulations across process corners.

2.3 Recommended Report Structure

You may adopt the following structure their design report to follow standard analogue IC design documentation practices:

Core Operational Amplifier Design

- Architecture & Schematic: Description and justification of the selected op-amp topology.
- Hand Calculations: Sizing equations for the differential input pair, active current mirrors, and any additional gain/output stages, where applicable.
- Simulation Results: DC gain, unity-gain bandwidth (GBW), and phase margin performance plots.

BGR Design

- Schematic and Operation: Explanation of the BGR core circuit (matching Q_1/Q_2 currents, PTAT/CTAT combination) and the implemented startup circuit.
- Performance Evaluation: Output reference voltage ($V_{ref} \approx 1.25$ V) and temperature sweep analysis across 0-75° C.

LDO Design

- Schematic and Operation: Integration of the error amplifier with pass transistor M_{pass} and feedback network.
- Stability and Transient Analysis: Load regulation, line regulation, and load transient response plots under dynamic current steps.

System Integration and Conclusion

- Cascaded PMU simulation results (noisy V_{DD} supply driving BGR $\rightarrow$ LDO $\rightarrow$ regulated output).
- Summary table comparing final simulated performance against initial specifications across process corners.

3 Automated Verification and Symbol Specifications

You are required to submit your Cadence design library as a single ZIP or TAR archive together with the technical report. The submitted design files must comply with the interface and submission requirements

specified in this section. Failure to provide a complete and compatible submission may result in marks being withheld under the relevant assessment criteria where functionality or performance cannot be verified.

The teaching team will assess each submission using a common Verilog-based testbench and simulation framework. A standard simulation environment will connect the testbench to the submitted transistor-level design using the course PDK models. All submissions will be evaluated under the same operating conditions and test procedures. The testbench will apply the specified supply and load conditions and automatically measure the relevant PMU performance metrics. Automated simulation results will provide the primary evidence for assessing specification compliance. Design methodology, engineering analysis, interpretation of results, and individual contributions will be assessed through the accompanying report.

Students are responsible for ensuring that their submissions are complete and compatible with the published interface requirements. Designs that cannot be simulated due to packaging, hierarchy, or interface errors may receive reduced marks where functionality or performance cannot be verified. Results presented in the report do not remove the requirement to submit a reproducible and verifiable design. All design file submissions must use the top-level interface in Table 3.

Table 3: PMU Symbol Specifications

Item	**Required Name**
Cadence design library	`cw_submission`
Top-level design cell	`pmu_top`
Required cell views	`schematic` and `symbol`
Netlist terminal order	`VDD` (left), `VSS` (bottom), `VOUT` (right)

The top-level symbol must contain the terminals specified in Table 4. Terminal names are case-sensitive. Students may choose their own internal circuit hierarchy and cell names, provided that the complete design remains accessible through the prescribed top-level interface. The teaching team will provide an interface template and a basic compatibility test prior to submission.

Table 4: PMU Terminal Specification

Terminal	**Cadence Pin Direction**	**Function**
`VDD`	`inputOutput`	External unregulated supply, driven by the testbench.
`VSS`	`inputOutput`	Ground and supply return.
`VOUT`	`output`	Regulated PMU output.

4 Marking Scheme

Table 5: Marking Scheme for Design Task 1

Section / Criteria	Detailed Description & Performance Indicators	Marks
Operational Amplifier & Sub-block Design & Sizing	**Hand Calculations:** Correct analytical derivation of transistor sizing, bias currents, and compensation components for the selected op-amp architecture. **Trade-off Justification:** Clear explanation of design trade-offs between power consumption, gain, bandwidth, and phase margin.	15
BGR Implementation	**Circuit Architecture:** Accurate implementation of the Bandgap Reference (BGR) core schematic utilizing the op-amp, bipolar transistors, and resistor networks as shown in the task diagram. **Startup & Stability:** Successful inclusion and verification of a startup circuit (addressing the design hint "How to start the bandgap?") and consideration of non-ideal effects such as channel-length modulation. **Performance Metrics:** Accurate generation of a stable reference voltage and demonstration of stability across the specified temperature range.	15
LDO Implementation	**Circuit Architecture:** Correct integration of the error amplifier with the power pass transistor (`M_P`) and feedback resistors (`R1`, `R2`) to regulate the unregulated input voltage to the required output voltage. **Regulation & Stability:** Proper sizing of the pass device to satisfy load current requirements and verification of loop stability through phase margin analysis.	15
Simulation Verification & Corner Analysis	Successful execution and presentation of DC, AC, and transient simulation results. Verification of performance across process corners (TT, FF, SS) and temperature sweeps.	10
Individual Contribution & Critical Reflection	Clear identification of individual student ownership within the pair (e.g., separating primary responsibility for the BGR, LDO, and system integration activities). Quality of critical reflection on design challenges encountered and solutions implemented during Task 1.	5
Total for Design Task 1		**60**

5 Design Task 2: Surface EMG Recording Amplifier Design

5.1 System Overview and Requirements

BM&M is developing a surface electromyography (sEMG) acquisition system for forearm gesture recognition using wet Ag/AgCl electrodes (See Figure 3). The system requires a front-end recording amplifier operating from a 1.8 V supply, providing approximately 40 dB midband gain and a bandwidth of 20-1000 Hz. The amplifier must accurately amplify low-level differential sEMG signals while achieving high common-mode rejection and low-noise operation [2]. The conditioned signals are digitised by an ADC and processed using machine-learning techniques for gesture recognition. A PMU, consisting of a BGR and a LDO, provides stable reference and supply voltages for the analogue front end (AFE).

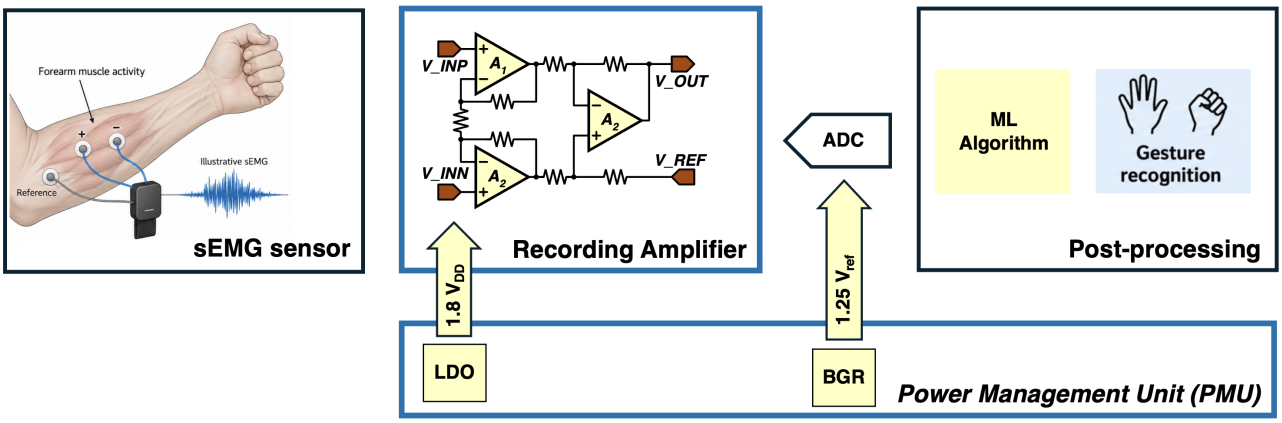


Figure 3: Overall system demonstration and Recording Amplifier design

Table 6: EMG Recording Amplifier Specifications

Parameter	**Basic Specification**
Supply voltage	1.8 V
Channel configuration	Differential input; single-ended output or fully differential output.
Voltage gain	40 dB; gain error $<$ 0.5 dB at 100 Hz.
Lower -3 dB cutoff	10 ± 2 Hz.
Upper -3 dB cutoff	1 ± 0.2 kHz.
Input-referred noise	$< 5\,\mu V_{RMS}$ integrated over 10–1000 Hz.
Linearity	THD $< 0.5\%$ at 100 Hz with a $2\,mV_{pp}$ differential input.
Differential electrode DC offset	Tolerate -50 mV to $+50$ mV after settling without saturation while retaining the specified gain and THD. This is the external electrode offset and not OTA input offset.
CMRR	$>$ 60 dB at 500 Hz in nominal matched simulation.
PSRR	$>$ 40 dB at 500 Hz.
Input impedance	≥ 10 MΩ at 500 Hz.
Channel current	$\leq 20\,\mu$A total quiescent current including local bias and common-mode circuitry; excludes the separately assessed LDO and BGR.
Stability and load	Phase margin $\geq 45^\circ$ if feedback is used.
Area	Report total capacitance and estimated device/passive area. No layout or fabricated-area requirement.

5.1.1 Recording Amplifier Architectures

Possible design approaches shown in Figure 4 include current-feedback instrumentation amplifiers (CFIAs) [3], capacitively coupled instrumentation amplifiers (CCIAs) [4], three-op-amp instrumentation amplifiers [5], and inverter-based amplifiers [6]. The references provided above are examples only. Students may select any suitable architecture. **Assessment will be based on engineering justification, design methodology, and demonstrated compliance with the specified performance requirements rather than the selected architecture.**

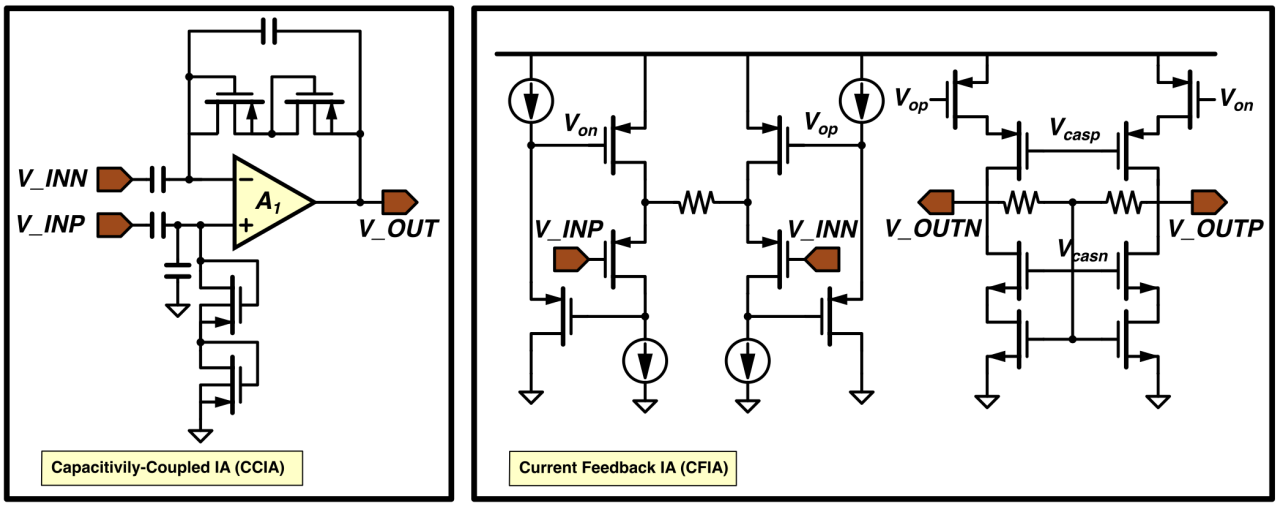


Figure 4: Example recording amplifier architectures. Students are encouraged to investigate alternative architectures and modify the example designs to achieve improved performance.

5.2 Collaborative Work Strategy for the Group

To efficiently design the sEMG recording amplifier, partners may divide responsibilities according to major functional blocks while jointly contributing to architecture selection, design calculations, and verification.

Group Member A (Input Stage & Noise Lead)

- Designs and sizes the differential input stage and associated bias circuitry.
- Investigates techniques for handling electrode DC offsets and common-mode interference.
- Performs small-signal analysis and hand calculations for gain, transconductance, and bandwidth.
- Conducts input-referred noise, offset, and CMRR simulations.
- Evaluates how transistor sizing and bias current affect noise, gain, and power consumption.

Group Member B (Output Stage & Stability Lead)

- Designs and sizes the gain/output stage, compensation network, and output buffer (where applicable).
- Implements the low-pass frequency response required to achieve the $1\,\mathrm{kHz}$ upper -3 dB cutoff frequency.
- Performs AC stability analysis, including phase margin and frequency response verification.
- Evaluates amplifier performance under different capacitive loads (1–5 pF) and process corners.
- Leads the preparation of Cadence symbols, design packaging, and automated verification compliance.

Joint Responsibilities

- Selection and justification of the overall amplifier architecture.
- Hand calculations for initial transistor dimensions and component values.
- Schematic capture and design reviews.
- Verification of all design specifications through DC, AC, transient, noise, and distortion simulations.
- Interpretation of results, discussion of design trade-offs, and preparation of the final report.
- Cross-checking simulation results and ensuring compatibility with the required symbol interface specification.

5.3 Report Expectations

Students are free to adopt a report structure that they believe best communicates their design methodology, analysis, and results. The report should demonstrate a logical engineering workflow and provide sufficient detail for the reader to understand and evaluate the design decisions made throughout the project. At a minimum, the report should contain:

- A clear description of the selected circuit architecture and its intended operation.
- Design methodology, including analytical calculations and transistor sizing rationale.
- Simulation-based verification of the specified performance requirements.
- Discussion of design trade-offs, limitations, and optimisation decisions.
- Critical evaluation of the final design against the target specifications.
- Clear identification of each student's individual contribution to the project.

The report will be assessed on both technical quality and the effectiveness with which engineering information is communicated. Students are therefore encouraged to organise the report professionally and present figures, schematics, tables, and simulation results clearly and concisely.

> Note: No design files are expected for Design Task 2. The evaluation will be based solely on the submitted report.

5.4 Marking Scheme for Design Task 2

Table 7: Marking Scheme for Design Task 2

Section / Criteria	Detailed Description & Performance Indicators	Marks
1. Amplifier Architecture Design & Sizing	**Hand Calculations:** Correct analytical derivation of transistor dimensions (W/L), bias currents, gain, bandwidth, and compensation components where applicable. **Architecture Justification:** Clear explanation of design choices and trade-offs between gain, noise, bandwidth, power consumption, and silicon area.	10
2. Recording Amplifier Implementation	**Circuit Architecture:** Correct implementation of the recording amplifier, including differential input stage, bias circuitry, gain stages, frequency-shaping circuitry, and output stage where applicable. **Performance Metrics:** Achievement of the required gain, bandwidth, CMRR, PSRR, and offset requirements.	10
3. Noise, Stability & Linearity Analysis	**Noise Performance:** Verification and discussion of input-referred noise and design choices used to minimise noise. **Stability Analysis:** Assessment of phase margin and operation under capacitive loads of 1–5 pF. **Linearity:** Evaluation of distortion performance and signal integrity.	8
4. Simulation Verification & Corner Analysis	Successful execution and presentation of DC, AC, transient, noise, and distortion simulations. Verification across process corners (TT, FF, SS), supply variations, and load capacitance conditions. Clear interpretation of results and comparison against specifications.	7
5. Individual Contribution & Critical Reflection	Clear identification of each student's ownership and technical contribution to the design (e.g., input-stage design, biasing network, output stage, stability optimisation, simulation campaign, automated verification preparation). Evidence of meaningful contribution to the project. Quality of critical reflection on design challenges, trade-offs, design iterations, and lessons learned throughout the development process.	5
Total for Design Task 2		**40**

References

[1] B. Razavi, "The Design of An LDO Regulator [The Analog Mind]," *IEEE Solid-State Circuits Magazine*, vol. 14, no. 2, pp. 7–17, Spring 2022, doi: 10.1109/MSSC.2022.3167308.

[2] X. Tang, Y. Liu, C. Lv, and D. Sun, "Hand Motion Classification Using a Multi-Channel Surface Electromyography Sensor," *Sensors*, vol. 12, no. 2, pp. 1130–1147, Jan. 2012, doi: 10.3390/s120201130.

[3] R. F. Yazicioglu, P. Merken, R. Puers, and C. Van Hoof, "A 60 μW 60 nV/$\sqrt{\text{Hz}}$ Readout Front-End for Portable Biopotential Acquisition Systems," *IEEE Journal of Solid-State Circuits*, vol. 42, no. 5, pp. 1100–1110, May 2007, doi: 10.1109/JSSC.2007.894804.

[4] Q. Fan, F. Sebastiano, J. H. Huijsing, and K. A. A. Makinwa, "A 1.8 μW 60 nV/$\sqrt{\text{Hz}}$ Capacitively-Coupled Chopper Instrumentation Amplifier in 65 nm CMOS for Wireless Sensor Nodes," *IEEE Journal of Solid-State Circuits*, vol. 46, no. 7, pp. 1534–1543, Jul. 2011, doi: 10.1109/JSSC.2011.2143610.

[5] Analog Devices, "MT-063: Signal-to-Noise Ratio (SNR), Signal-to-Noise-and-Distortion Ratio (SINAD), and Effective Number of Bits (ENOB)," Available: https://www.analog.com/media/en/training-seminars/tutorials/MT-063.pdf [Accessed: Oct. 1, 2026].

[6] R. Vieira, F. Näf, R. Martins, N. Horta, N. Lourenço, and R. Póvoa, "A Tunable Gain and Bandwidth Low-Noise Amplifier with 1.44 NEF for EMG and EOG Biopotential Signal," *Electronics*, vol. 12, no. 12, p. 2592, Jun. 2023, doi: 10.3390/electronics12122592.

A The 20-Point Marking Scale Assessment Criteria

This coursework is assessed using the University of Bristol 20-point marking scale [2], which is subsequently mapped to the institutional 100-point scale for the purposes of unit mark calculation and degree classification. The criteria defined below provide guidance on the expected standard of performance across key aspects of the submission, including achievement of design specifications, accuracy of analytical calculations, quality of Cadence Virtuoso implementation and simulation results, and the depth of engineering analysis. Higher marks are awarded for designs that demonstrate strong technical competence, rigorous validation, clear presentation, and critical evaluation of design trade-offs, while lower marks reflect deficiencies in functionality, supporting calculations, simulation evidence, and engineering understanding.

The descriptors in Table 8 are indicative rather than exhaustive, and marks will be awarded using academic judgement, taking into account the overall quality, originality, technical depth, and coherence of the submission. Consequently, not every criterion within a band must be satisfied in full for a particular mark to be awarded, and exceptional strengths or significant weaknesses in one area may influence the final mark within the appropriate grade band.

Table 8: Marking Scale

Band	**Points (100-pt Equivalent)**	**Assessment Criteria & Performance Indicators**
Excellent	20 (100), 19 (94), 18 (83), 17 (78)	**Specifications:** All target specifications comfortably met or exceeded. **Calculations:** Hand calculations match Cadence Virtuoso simulation outputs with negligible error. **Presentation:** Virtuoso schematics are professionally organized; waveforms are impeccably presented with clear markers. **Analysis:** Deep, highly sophisticated critical analysis of engineering trade-offs.
Very Good / Good	16 (75), 15 (72), 14 (68), 13 (65)	**Specifications:** Most design specifications successfully met; minor deviations justified logically. **Calculations:** Analytical sizing calculations are accurate, though minor tuning was needed in simulation. **Presentation:** Correct use of Cadence ADE for DC, AC, and Transient analyses with clean, readable data plots. **Analysis:** Solid understanding of design trade-offs and structural loop stability.
Satisfactory	12 (62), 11 (58), 10 (55), 9 (52)	**Specifications:** Circuit operational framework functions, but fails to meet multiple key specifications. **Calculations:** Hand calculations contain minor mathematical or parameter-mapping errors. **Presentation:** Cadence Virtuoso simulations are present, but missing key node data or proper plot formatting. **Analysis:** Descriptive rather than critical reflection on design trade-offs.
Marginal Pass / Fail	8 (48) – 1 (7)	**Specifications:** Design fails to function; transistors are incorrectly biased in triode/cutoff. **Calculations:** Severe omissions in hand calculations or complete lack of circuit sizing methodology. **Presentation:** Simulation data is incomplete, flawed, or absent from the report. **Analysis:** Little to no understanding of internal circuit dynamics, stability, or power allocation.

[2] https://www.bristol.ac.uk/academic-quality/assessment/regulations-and-code-of-practice-for-taught-programmes/marking-criteria/